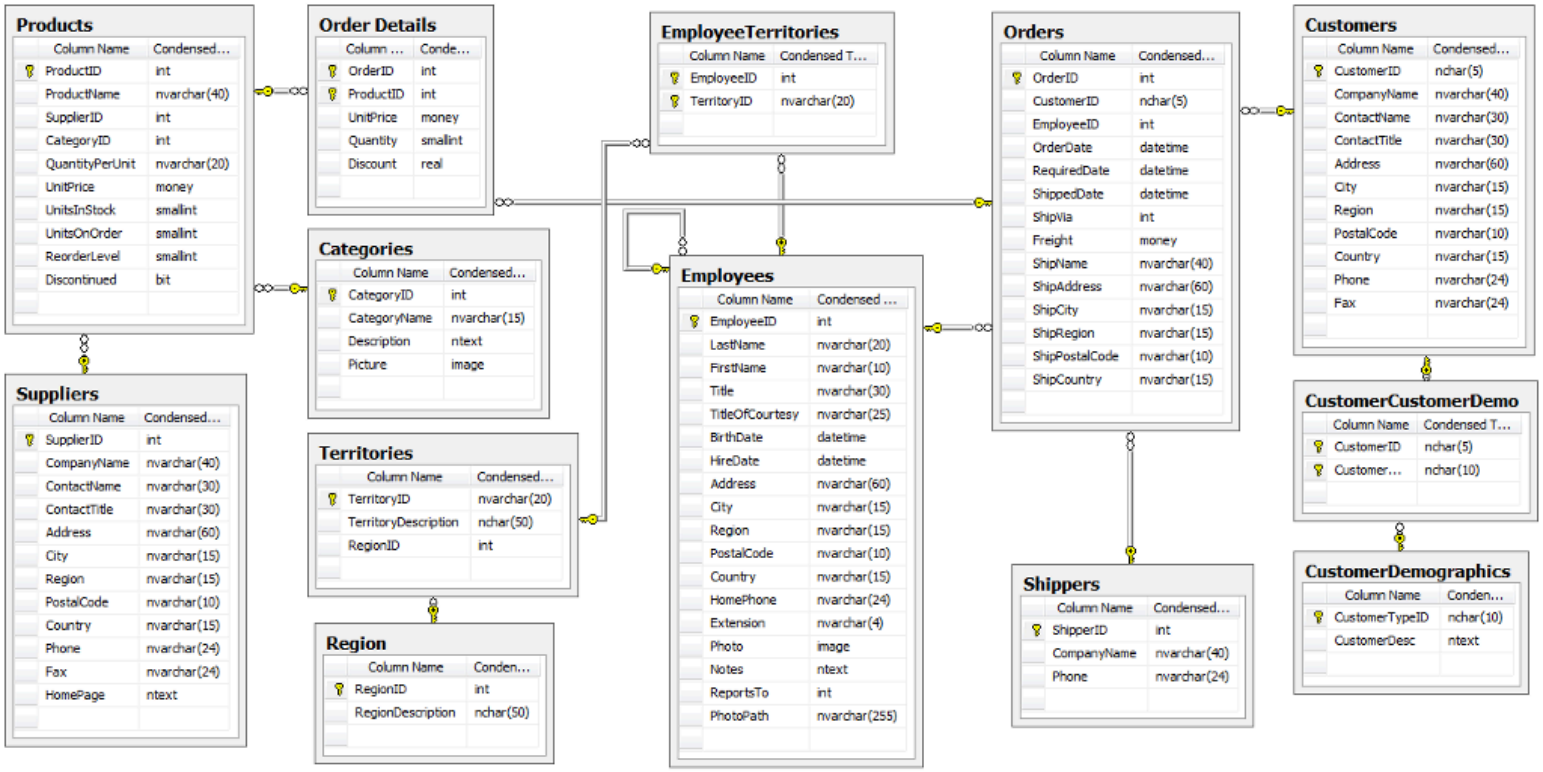
Lab Task #08

Structured Query Language (SQL)

Given the query file provide to create Northwind database schema solve following queries using NORTHWIND Database:

(solve all queries using joins only)

1. Provide the order ids of all orders that are shipped using 'Federal Shipping'.
2. Select the name, address, city, and region of Customers that have placed orders to be delivered in Madrid.
3. Extract the ids and the names in upper case for customers who haven't placed any orders yet.
4. Find the total number of products ordered from each supplier, including suppliers who have not supplied any products.
5. Provide the names of the customers and their cities who belong to the same city as the employee who processed their order. Also display the full names of the employees.
6. Display the employees (managers) who manage at least 3 employees.
7. Fetch top 5 most popular products, products that have the maximum quantity of units sold.
8. Create a view to find the total number of products ordered by each customer.
9. Find all products and their corresponding supplier names, but only return products that have not been discontinued.
10. Retrieve the most recent order (by order date) placed by each customer.
11. Identify employees who have processed more than 10 orders for any individual customer.
12. List all products where the total quantity ordered exceeds 100 units.
13. Find suppliers that have supplied products to orders shipped by a specific shipper. (e.g., ShipperID = 1)
14. Find employees who have handled orders for customers in the same country as the employee.
15. Create a view to retrieve the most expensive product from each category.


Products
Column Name
Condensed...
ProductID
int
ProductName
nvarchar(40)
SupplierID
int
CategoryID
int
QuantityPerUnit
nvarchar(20)
UnitPrice
money
UnitsInStock
smallint
UnitsOnOrder
smallint
ReorderLevel
smallint
Discontinued
bit
Order Details
Column ...
Conde...
OrderID
int
ProductID
int
UnitPrice
money
Quantity
smallint
Discount
real
EmployeeTerritories
Column Name
Condensed T...
EmployeeID
int
TerritoryID
nvarchar(20)
Orders
Column Name
Condensed...
OrderID
int
CustomerID
nchar(5)
EmployeeID
int
OrderDate
datetime
RequiredDate
datetime
ShippedDate
datetime
ShipVia
int
Freight
money
ShipName
nvarchar(40)
ShipAddress
nvarchar(60)
ShipCity
nvarchar(15)
ShipRegion
nvarchar(15)
ShipPostalCode
nvarchar(10)
ShipCountry
nvarchar(15)
Customers
Column Name
Condensed...
CustomerID
nchar(5)
CompanyName
nvarchar(40)
ContactName
nvarchar(30)
ContactTitle
nvarchar(30)
Address
nvarchar(60)
City
nvarchar(15)
Region
nvarchar(15)
PostalCode
nvarchar(10)
Country
nvarchar(15)
Phone
nvarchar(24)
Fax
nvarchar(24)
Categories
Column Name
Condensed...
CategoryID
int
CategoryName
nvarchar(15)
Description
ntext
Picture
image
Employees
Column Name
Condensed ...
EmployeeID
int
LastName
nvarchar(20)
FirstName
nvarchar(10)
Title
nvarchar(30)
TitleOfCourtesy
nvarchar(25)
BirthDate
datetime
HireDate
datetime
Address
nvarchar(60)
City
nvarchar(15)
Region
nvarchar(15)
PostalCode
nvarchar(10)
Country
nvarchar(15)
HomePhone
nvarchar(24)
Extension
nvarchar(4)
Photo
image
Notes
ntext
ReportsTo
int
PhotoPath
nvarchar(255)
CustomerCustomerDemo
Column Name
Condensed T...
CustomerID
nchar(5)
Customer...
nchar(10)
Suppliers
Column Name
Condensed...
SupplierID
int
CompanyName
nvarchar(40)
ContactName
nvarchar(30)
ContactTitle
nvarchar(30)
Address
nvarchar(60)
City
nvarchar(15)
Region
nvarchar(15)
PostalCode
nvarchar(10)
Country
nvarchar(15)
Phone
nvarchar(24)
Fax
nvarchar(24)
HomePage
ntext
Territories
Column Name
Condensed...
TerritoryID
nvarchar(20)
TerritoryDescription
nchar(50)
RegionID
int
Shippers
Column Name
Condensed...
ShipperID
int
CompanyName
nvarchar(40)
Phone
nvarchar(24)
CustomerDemographics
Column Name
Conden...
CustomerTypeID
nchar(10)
CustomerDesc
ntext
Region
Column Name
Conden...
RegionID
int
RegionDescription
nchar(50)